

Semantic News and Oil Price Database Project Report

Prepared: May 6, 2026

Workspace: DatabaseProject

Database target: MySQL

Modeling framework: scikit-learn

Executive Summary

This project combines semantic/geopolitical news data with oil market pricing data to support both relational analysis and predictive oil price modeling. The workspace datasets were organized into a MySQL-ready schema with operational source tables, dimensional reporting tables, and analysis views. A scikit-learn model was trained to predict the next trading-day Brent crude oil price using market, lag, volatility, event, and geopolitical risk features.

The final model uses a `StandardScaler` -> `Ridge Regression` pipeline. On a chronological test split covering December 22, 2022 through March 12, 2026, the model achieved a test RMSE of **1.516 USD** and MAE of **1.118 USD**. The previous-price baseline RMSE was **1.536 USD**, so the fitted model performs slightly better than a strong short-horizon benchmark.

Project Objectives

- Compile all supplied CSV datasets into a MySQL database.
- Preserve both operational data and dimensional/fact reporting structures.
- Create analysis-ready SQL views for daily oil/news features, event reaction analysis, and country petrol impact analysis.
- Train and export a predictive model for next trading-day Brent oil pricing.
- Produce reusable artifacts for loading, modeling, prediction, and reporting.

Data Sources and Coverage

The workspace contains 20 CSV files under `datasets/`. The most important modeling and database tables are:

Dataset	Rows	Purpose
<code>ops_market_daily.csv</code>	4,047	Daily Brent/WTI prices, macro-market indicators, lag features, volatility, and event flags
<code>ops_gpr_daily.csv</code>	15,078	Daily geopolitical risk/news indices and article counts
<code>ops_gpr_monthly.csv</code>	1,515	Monthly geopolitical risk indicators
<code>ops_gpr_country_monthly.csv</code>	66,660	Country-month geopolitical risk measures
<code>ops_events.csv</code>	55	Named geopolitical, war, sanctions, disaster, and market events
<code>ops_countries.csv</code>	18	Country dimension for impact and petrol price analysis

Dataset	Rows	Purpose
ops_petrol_price_snapshots.csv	28	Country petrol price snapshots and price change indicators
dim_* / fact_* files	mixed	Warehouse-ready star schema tables

The main modeling table, `ops_market_daily`, spans **February 17, 2010 through March 12, 2026**. Daily GPR data spans **January 1, 1985 through April 13, 2026**.

Database Design

The database design uses two complementary layers.

Operational Layer

The operational layer directly represents the supplied source-style datasets. It is useful for model training, exploration, and tracing records back to the original CSVs.

Key tables:

- `ops_market_daily`: primary daily modeling table with oil prices, lagged oil prices, market indicators, volatility metrics, and event fields.
- `ops_gpr_daily`: daily semantic/geopolitical news risk features.
- `ops_events`: named events with event type, category, description, date, and severity.
- `ops_countries`, `ops_country_impact`, and `ops_petrol_price_snapshots`: country exposure and petrol price impact analysis.
- `ops_gpr_country_monthly`: country-month geopolitical risk history.

Important joins:

- `ops_market_daily.market_date = ops_gpr_daily.gpr_date`
- `ops_market_daily.market_date = ops_events.event_date`
- `ops_market_daily.market_date = ops_crude_oil_daily.trade_date`
- `ops_countries.country_id = ops_country_impact.country_id`
- `ops_countries.country_id = ops_petrol_price_snapshots.country_id`
- `ops_countries.iso3 = ops_gpr_country_monthly.iso3`

Dimensional Layer

The dimensional layer supports BI-style reporting with reusable dimensions and facts.

Key dimensions:

- `dim_date`
- `dim_country`
- `dim_event`

Key facts:

- `fact_market_daily`
- `fact_gpr_daily`
- `fact_gpr_monthly`

- fact_country_impact
- fact_petrol_prices

This layout enables date-based time-series reporting, country impact comparison, and event-linked price analysis.

SQL Views

The project creates three analysis views in `sql/analytics_views.sql`.

View	Description
<code>vw_daily_oil_news_features</code>	Daily joined Brent/WTI, GPR, event, and volatility features for analysis and modeling
<code>vw_event_price_reaction</code>	Event-date oil price and risk indicators for event study analysis
<code>vw_country_petrol_impact</code>	Country exposure, petrol price snapshots, and vulnerability indicators

Example analysis query:

```
SELECT market_date, brent_price_usd, gpr_index, event_type, event_severity
FROM vw_daily_oil_news_features
WHERE event_flag = 1
ORDER BY market_date DESC
LIMIT 20;
```

MySQL Loading Workflow

The MySQL loader is implemented in `src/load_mysql.py`. It:

- Reads all CSV files from `datasets/`.
- Uses `datasets/data_dictionary.csv` to infer MySQL column types.
- Creates one table per CSV file.
- Batches inserts for efficient loading.
- Applies the SQL analysis views after loading.

Recommended load command:

```
python src\load_mysql.py --replace
```

The optional `docker-compose.yml` starts a local MySQL 8.4 container for the project.

Predictive Modeling Approach

The predictive target is:

Next trading-day Brent crude oil price in USD

The training script is `src/train_oil_model.py`. It uses a chronological train/test split, which is appropriate for time-series style financial data because future rows are not allowed to influence past training rows.

Model pipeline:

```
StandardScaler -> Ridge Regression
```

Ridge regression was selected because it is transparent, fast, reproducible, and suitable for a database project showcase. Standardization helps keep the regularization behavior consistent across features with different units.

Model Features

The model uses 20 features from `ops_market_daily.csv`:

Feature Group	Columns
Current oil prices	brent_price_usd, wti_price_usd
Market indicators	dxy_index, vix_index
News/risk indicator	gpr_index
Returns	brent_return, wti_return
Lagged prices	brent_lag_1, brent_lag_3, brent_lag_7, wti_lag_1, wti_lag_3, wti_lag_7
Volatility	brent_volatility_7d, brent_volatility_30d, wti_volatility_7d, wti_volatility_30d
Spread	brent_wti_spread
Event features	event_severity, event_flag

Model Results

Training metadata:

Item	Value
Training rows	3,236
Test rows	810
Training date range	2010-02-18 to 2022-12-21
Test date range	2022-12-22 to 2026-03-12
Ridge alpha	0.1

Performance:

Metric	Training	Test	Previous-Price Baseline
MAE	1.080	1.118	1.113
RMSE	1.567	1.516	1.536
MAPE	1.515%	1.463%	1.457%
R-squared	0.996	0.967	0.966

The model narrowly improves RMSE versus the previous-price baseline. This is a realistic outcome for next-day oil price prediction, where yesterday's price is usually a very strong benchmark. The model's value is that it gives a reproducible framework for incorporating market, event, and semantic news features while remaining explainable.

Latest example prediction:

Input Date	Current Brent	Predicted Next Trading-Day Brent
2026-03-12	95.76 USD	95.90 USD

Generated Artifacts

Artifact	Purpose
model_artifacts/oil_price_model.joblib	Fitted scikit-learn pipeline
model_artifacts/oil_price_model.json	Model metadata, metrics, feature names, and date ranges
model_artifacts/test_predictions.csv	Test-set predictions against actual Brent prices
docs/database_schema.md	Mermaid ERD and schema notes
sql/analytics_views.sql	SQL views for joined analysis
reports/oil_news_database_model_report.md	Markdown version of this report
output/pdf/oil_news_database_model_report.pdf	PDF version of this report

Recommended Next Steps

- Add explicit foreign key constraints after confirming load order and null behavior.
- Add indexes on dates and join keys such as `market_date`, `gpr_date`, `event_date`, `country_id`, and `iso3`.
- Add a notebook or dashboard that reads from `vw_daily_oil_news_features`.
- Compare Ridge against tree-based scikit-learn models such as `RandomForestRegressor` or `HistGradientBoostingRegressor`.
- Add rolling-window validation to better evaluate model stability over time.
- Store model predictions back into MySQL in a `model_predictions` table for dashboarding.

Conclusion

The completed project demonstrates a full data workflow: source CSVs are transformed into a MySQL-ready analytical database, semantic news and geopolitical risk signals are joined with oil market features, and a scikit-learn predictive model is trained and exported. The result is suitable for a database project presentation because it includes schema design, SQL loading, analysis views, reproducible modeling, and report-ready artifacts.